ECOllateral: A2 team contributions

CSCI 4150 | Mid-Semester System & Evidence Review | Shaun, Troy, Scott

We divided the project into three complementary areas: system development, environmental data and retrieval, and evaluation and communication. Each member's contribution connects to the prototype and evidence presented in our A2 review.

Shaun: system development and integration

Shaun developed the system architecture and connected the environmental inputs, prediction models, physical checks, and permit results through the API. He handled model training and the RandomForest/XGBoost comparison, documenting why XGBoost was selected for the prototype. His work established the path from a facility request to a daily cooling-water estimate, prediction range, and warnings.

Evidence: system map, API and model implementation, and E1 model-comparison results.

Troy: environmental data and permit retrieval

Troy worked on environmental data acquisition and preparation, including station and climate-division checks and drought and groundwater interpretation. He analyzed missing-data handling and the biased-station failure in E2. He also worked on permit retrieval and geographic filtering, connecting exact quotations to the requested location and documenting the fictional location tests in E4.

Evidence: data contracts, E2 missingness results and donor diagnostic, and E4 retrieval cases.

Scott: evaluation and evidence presentation

Scott evaluated the physical-rule checks in E3 and prediction-range coverage in E5. He analyzed what the failures mean for the design, including the need for separate calibration data and further testing with measured consumption. He organized the evidence, slides, and presentation materials to explain the results, design decisions, and limits of the current prototype.

Evidence: E3 constraint cases, E5 coverage results, evidence summary, dossier, and slides.

Presentation responsibilities

- **Shaun - slides 1-2:** problem, intended users, scope, and system architecture.
- **Troy - slides 3-4:** model comparison, missing readings, and the biased-station failure.
- **Scott - slides 5-6:** prediction ranges, evidence-based decisions, and next steps.

Each member presents two slides. The talk targets four minutes, followed by questions. The current results use synthetic data; testing at real facilities and assessing broader regional water impacts remain future work.